Orange – U4 | PES-VCS report

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4th Sem E-Sec

Repository : https://github.com/nandan-chiru/PES2UG24CS301-pes-vcs.git

```
nandan_c@Anamika: cd PES2UG24CS301-pes-vcs/
nandan_c@Anamika:~/PES2UG24CS301-pes-vcs$ make # Build the pes binary
make all # Build pes + test binaries
make clean # Remove all build artifacts
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
make: Warning: File 'index.o' has modification time 2.2 s in the future
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
make: warning: Clock skew detected. Your build may be incomplete.
rm -f pes test objects test tree object.o tree.o index.o commit.o pes.o test objects.o
test tree.o
rm -rf .pes
nandan_c@Anamika:~/PES2UG24CS301-pes-vcs$ export PES_AUTHOR="NANDAN C <PES2UG24CS301>"
nandan_c@Anamika:~/PES2UG24CS301-pes-vcs$ make test objects
./test objects
gcc -Wall -Wextra -O2 -c test objects.c -o test objects.o
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -o test_objects test_objects.o object.o -lcrypto
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at:
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check
All Phase 1 tests passed.
nandan_c@Anamika:~/PES2UG24CS301-pes-vcs$ find .pes/objects -type f
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
```

Phase – 2 : Tree objects

```
nandan_c@Anamika:~/PES2UG24CS301-pes-vcs$ make test tree
gcc -Wall -Wextra -O2 -c test tree.c -o test tree.o
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -o test tree test tree.o object.o tree.o -lcrypto
nandan_c@Anamika:~/PES2UG24CS301-pes-vcs$ ./test tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization
$ find .pes/objects -type f
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
$ xxd .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12 | head
-20
00000000: 626c 6f62 2031 3600 4865 6c6c 6f2c 2050  blob 16.Hello, P
00000010: 4553 2d56 4353 210a                      ES-VCS!.
```

Phase – 3 : The index file

```
nandan_c@Anamika:~/PES2UG24CS301-pes-vcs$ make pes
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
nandan_c@Anamika:~/PES2UG24CS301-pes-vcs$ make pes
./pes init
echo "hello" > file1.txt
echo "world" > file2.txt
./pes add file1.txt file2.txt
./pes status
make: 'pes' is up to date.
Initialized empty PES repository in .pes/
Staged changes:
 staged: file1.txt
 staged: file2.txt
Unstaged changes:
 (nothing to show)
Untracked files:
 untracked: test_tree.c
 untracked: pes.c
 untracked: tree.h
 untracked: Makefile
 untracked: pes.h
 untracked: index.h
 untracked: test_objects.c
 untracked: commit.h
 untracked: index.c
 untracked: README.md
 untracked: test_tree
 untracked: test_sequence.sh
 untracked: test_objects
 untracked: commit.c
 untracked: object.c
 untracked: tree.c
$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776594383 6
file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776594387 6
file2.txt
```

Phase 4: Commits and History

```
nandan_c@Anamika:~/PES2UG24CS301-pes-vcs$ ./pes init
echo "Hello" > hello.txt
./pes add hello.txt
./pes commit -m "Initial commit"
echo "World" >> hello.txt
./pes add hello.txt
./pes commit -m "Add world"
echo "Goodbye" > bye.txt
./pes add bye.txt
./pes commit -m "Add farewell"
./pes log
Initialized empty PES repository in .pes/
Committed: 1298c3f805d8... Initial commit
Committed: 2d2c3fac0685... Add world
Committed: 7b143f848131... Add farewell
commit 7b143f848131e298bef1c833005a0c8b69bd8c99dcd13f6eb53eaf62586f911f
Author: NANDAN C <PES2UG24CS301>
Date:   1776594545
 Add farewell

commit 2d2c3fac06858ebbacf7ec7964ec38058208eff5e90d8a578ea1e41c67420ae4
Author: NANDAN C <PES2UG24CS301>
Date:   1776594545
 Add world

commit 1298c3f805d8055e5afca262cff30f45a9f57b9d4bb0bf8384bb9bec7553a17f
Author: NANDAN C <PES2UG24CS301>
Date:   1776594545
 Initial commit
nandan_c@Anamika:~/PES2UG24CS301-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/12/98c3f805d8055e5afca262cff30f45a9f57b9d4bb0bf8384bb9bec7553a17f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
```

```
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/2d/2c3fac06858ebbacf7ec7964ec38058208eff5e90d8a578ea1e41c67420ae4
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/7b/143f848131e298bef1c833005a0c8b69bd8c99dcd13f6eb53eaf62586f911f
.pes/objects/85/e367b76c2417b47a95f46208835ad8700dd58335171c94a497974be6191900
.pes/objects/96/c0d5c28dd3b719606c36723cd16aefbc73b3dc8b74de61070183357d6968b9
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/objects/f1/0239b620372163c2dd06ac33189180c55edf1c56b15990a1cf44e00b1b0e59
.pes/refs/heads/main
nandan_c@Anamika:~/PES2UG24CS301-pes-vcs$ cat .pes/refs/heads/main
cat .pes/HEAD
7b143f848131e298bef1c833005a0c8b69bd8c99dcd13f6eb53eaf62586f911f
ref: refs/heads/main
```

FINAL TEST

```
nandan_c@Anamika:~/PES2UG24CS301-pes-vcs$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===
--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
 staged: file.txt
 staged: hello.txt
Unstaged changes:
 (nothing to show)
Untracked files:
 (nothing to show)

--- First Commit ---
Committed: 1c2e3826157c... Initial commit
Log after first commit:
commit 1c2e3826157cf3e1f1ebdff557135c65d2356b8bbe15b326403224731c0dcc55
Author: NANDAN C <PES2UG24CS301>
Date:   1776594666
 Initial commit

--- Second Commit ---
Committed: ceba5f02fe47... Update file.txt

--- Third Commit ---
Committed: 70799d0e4a41... Add farewell

--- Full History ---
commit 70799d0e4a4131432a6f5444895c95e0236fd786857059748be7cd71566233dd
Author: NANDAN C <PES2UG24CS301>
Date:   1776594666
 Add farewell

commit ceba5f02fe4745e82ebf2c737f3f10c66a5876fc6f71c1efd1c23684db396663
Author: NANDAN C <PES2UG24CS301>
Date:   1776594666
 Update file.txt

commit 1c2e3826157cf3e1f1ebdff557135c65d2356b8bbe15b326403224731c0dcc55
Author: NANDAN C <PES2UG24CS301>
Date:   1776594666
 Initial commit

--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
70799d0e4a4131432a6f5444895c95e0236fd786857059748be7cd71566233dd
```

```
--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/1c/2e3826157cf3e1f1ebdff557135c65d2356b8bbe15b326403224731c0dcc55
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/70/799d0e4a4131432a6f5444895c95e0236fd786857059748be7cd71566233dd
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/ce/ba5f02fe4745e82ebf2c737f3f10c66a5876fc6f71c1efd1c23684db396663
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

=== All integration tests completed ===
```

Phase – 5,6

Q5.1) Checkout Implementation

When pes checkout <branch> is executed, the system switches from the current branch to the target branch.

First, the HEAD is updated to point to the new branch. If the branch does not exist, a new reference file is created inside the .pes/refs/heads/ directory.

Next, the working directory is updated to match the target branch:

- *Files present only in the target branch are created*
- *Files present only in the current branch are deleted*
- *Files that differ are overwritten*

Before making any changes, the system must check for conflicts:

- *If a file has local uncommitted changes and also differs in the target branch → checkout is blocked*
- *If an untracked file would be overwritten → checkout is blocked*

Finally, any empty directories left after file deletions are removed.

Q5.2) Dirty Working Directory Detection

To detect if a file is modified (dirty):

- *For each file in the index:*
  - *Compare its mtime and size with the working directory file*
  - *If they differ: Recompute the file's hash. Compare with the stored blob hash.*

Then compare with the target branch:

- *Get the file's hash from the target branch's tree*

Decision:

- *If file is modified locally AND different in target branch → checkout fails*
- *If only one condition is true → safe to proceed*

Q5.3) Detached HEAD

- *In a detached HEAD state, HEAD points directly to a commit hash, not a branch.*

- *New commits are still created*
- *But no branch is updated to track them*
- *Problem: These commits become unreachable when you switch branches and may be deleted later.*

Q6.1) GC Algorithm and Object Visit Estimate

The Garbage Collector uses a Mark-and-Sweep approach to maintain repository health. In the Mark phase, the algorithm identifies reachable objects by traversing the directed acyclic graph (DAG) starting from HEAD and all branch references. Each encountered commit enqueues its tree and parent hashes, while trees enqueue their respective blobs and subtrees into a processing queue. To ensure efficiency and prevent redundant processing, a hash set tracks all visited objects, providing O(1) lookup times. Once the queue is empty, the Sweep phase begins by iterating through every file in the .pes/objects directory. Any file whose hash is not present in the "reachable" set is deemed an orphan and deleted from the disk. Finally, empty shard directories are removed to keep the file system clean. For a scale of 100,000 commits, deduplication typically limits visits to 500,000 objects, though unique content could spike this to millions.

Q6.2) GC Race Condition

A race condition occurs when the GC executes simultaneously with a new commit operation. During a commit, a blob may be written to the object store before the parent tree or commit object is finalized and linked to a reference. If the GC's Mark phase runs at this precise moment, it will fail to find a path from any ref to that new blob, marking it as unreachable. If the commit then finishes its work, the repository temporarily looks valid, but the subsequent Sweep phase will delete the "unreachable" blob. This results in a corrupted repository where a valid commit points to a non-existent object. To prevent this, the system implements a grace period, typically skipping any objects created within a recent time threshold (like 30 minutes or two weeks). Additionally, lockfiles are used during reference updates to ensure that the GC backs off when it detects an active write operation in progress.